Project 1 – Digital Forensics

Setup a Forensic Lab Environment, the Computer Aided Investigative Environment (CAINE) and Image a Hard Disk Using Different Tools

Introduction

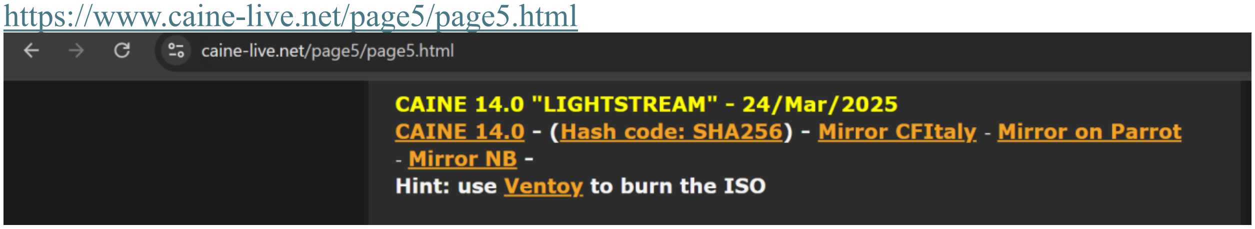
Digital forensic roles have different phases and mainly they need to focus on properly identifying, preserving, acquiring, and analyzing digital evidence. The integrity of digital evidence has to be maintained throughout the investigation process. This lab is designed to set up a forensic environment using Computer Aided INvestigative Environment (CAINE) open-source Linux live distribution (operating system) within Oracle VirtualBox to perform forensic imaging of an attached evidence disk.

The lab uses two different digital forensic tools called Guymager and dcfldd to acquire forensic image. The lab focused on mounting the evidence disk in read-only mode to maintain integrity and also cryptographic hash values were generated to verify the integrity of acquired images from both tools. By using different tools to acquire same evidence imaging provides practical differences between forensic image types and importance of hashing to maintain the data integrity.

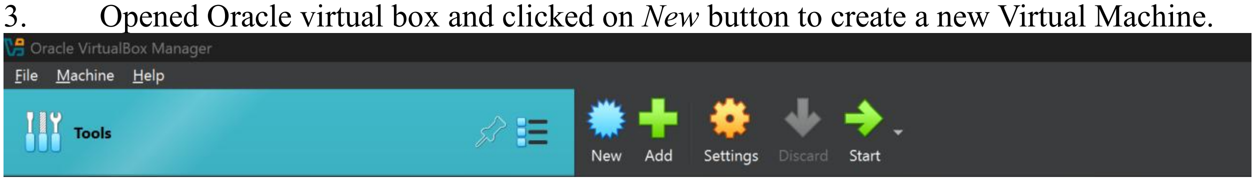
Part 1- Setting up CAINE on Oracle Virtual box

1. Logged to Azure Lab environment using following web link. https://labs.azure.com/virtualmachines?feature_vnext=true

2. Downloaded the iso file for Caine 14.0 using URL https://www.caine-live.net/page5/page5.html

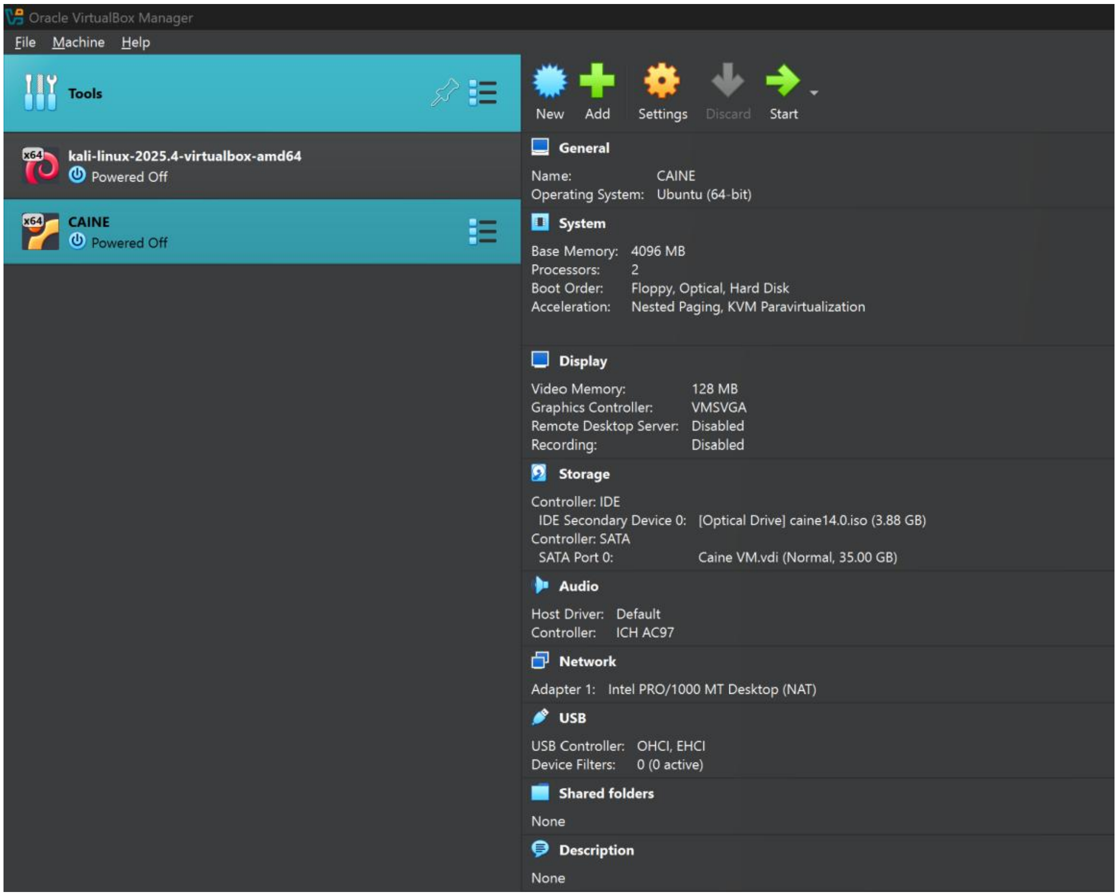


3. Opened Oracle virtual box and clicked on *New* button to create a new Virtual Machine.



4. Created a VM with following settings.

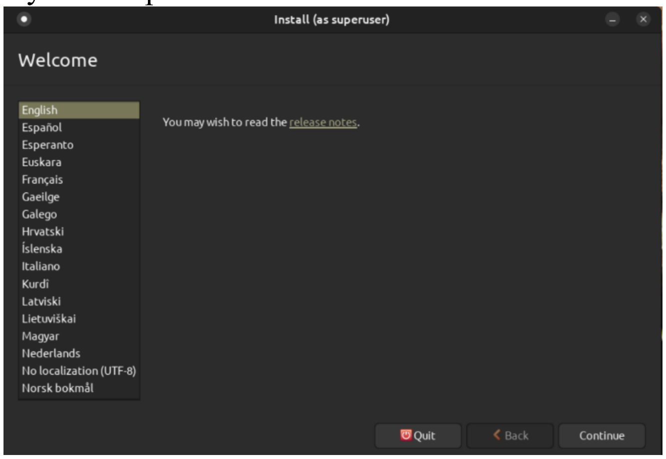
   VM Name – CAINE
   Operating System – Type: Linux and Version: Ubuntu (64-bit)
   System – 4GB RAM and 2 CPU cores
   Display – Video memory 128 MB
   Storage – Hard Disk 35 GB
   Attach CAINE ISO as optical disk

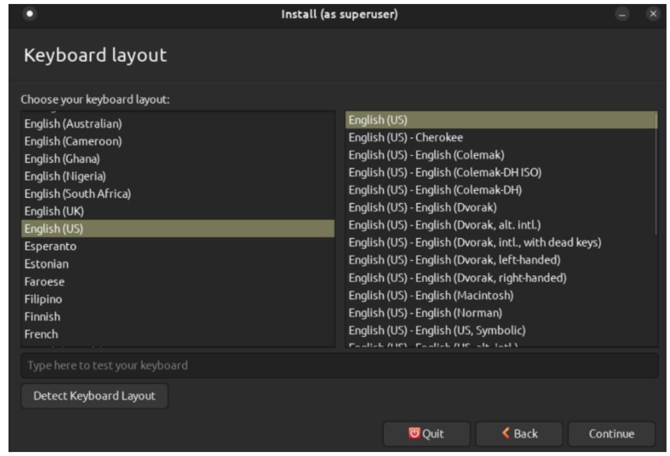


Part 2- Installing CAINE OS within the Caine VM

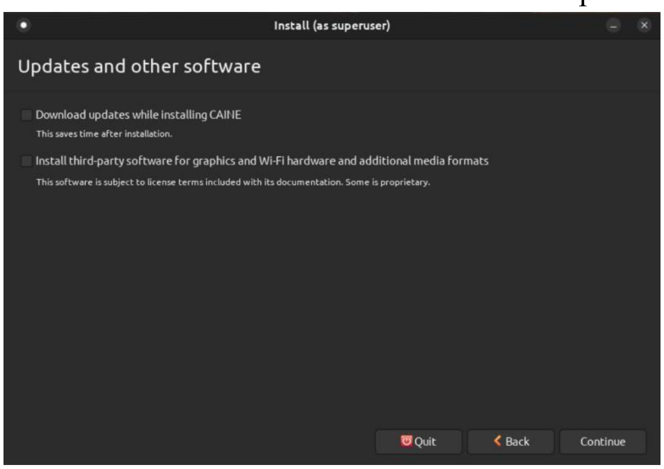
1. Started the Caine VM.

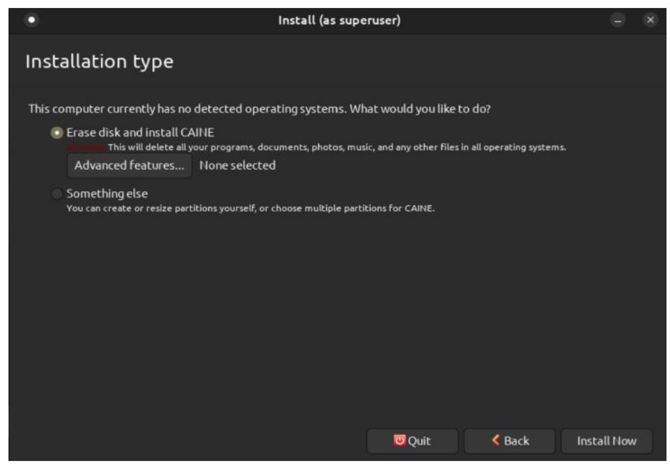
2. On the desktop, located ‘Install Caine24.04’ and clicked on the icon to start installation. Then selected ‘English’ as language and pressed ‘Continue’. Selected ‘English (US)’ as keyboard layout and pressed ‘Continue’.



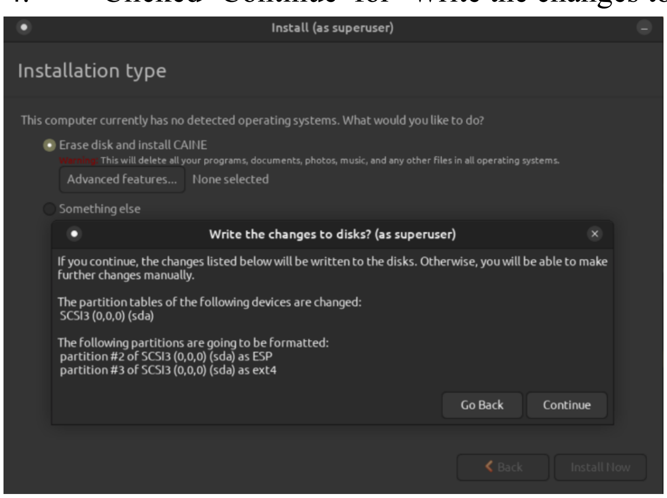


3. Unchecked ‘Download updates…’, and ‘Install third-party…’ and pressed ‘Continue’. Selected ‘Erase disk and install CAINE’ and pressed ‘Install Now’.

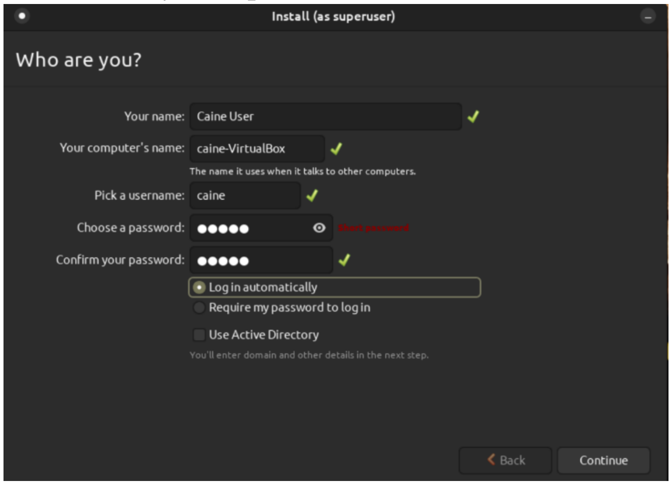




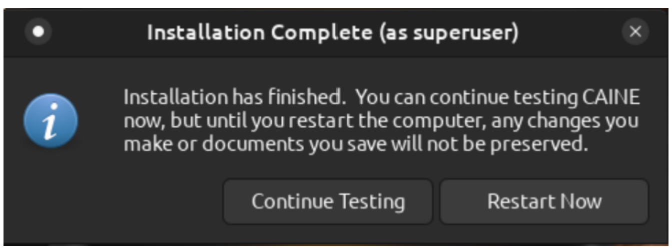
4. Clicked ‘Continue’ for ‘Write the changes to disk? and selected timezone as Chicago.



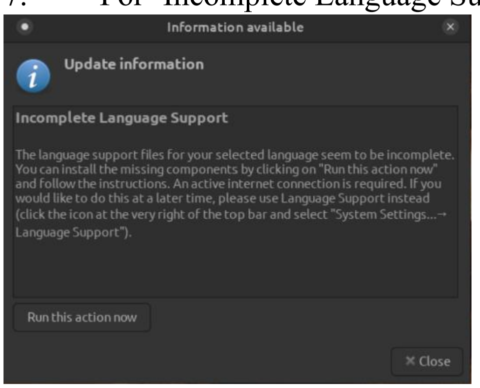
5. Entered 'Caine User' for Who are you, and 'Caine' as a password. Then select 'Log in automatically' and pressed 'Continue'. CAINE OS was started to instrall.

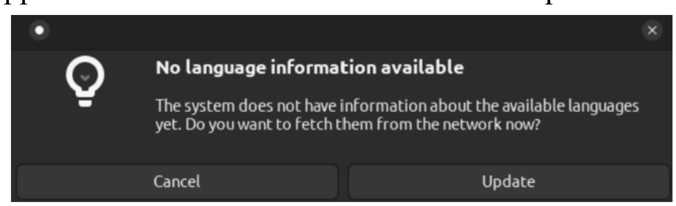


6. When installation completed, select 'Restart Now'. Pressed enter when asked to remove installation medium.

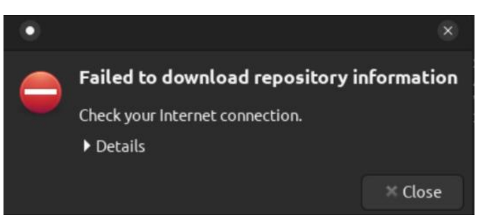


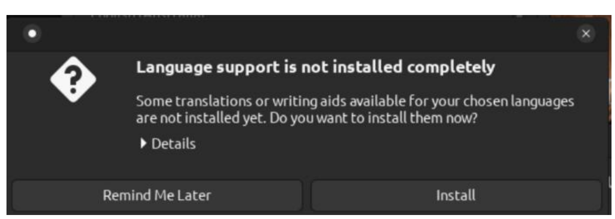
7. For 'Incomplete Language Support' clicked on 'Run this action now'. Update.





8. Closed error message that appears for 'Noble' and clicked 'Install' for 'Language Support is not Installed Completely'. Clicked 'Close' when complete.

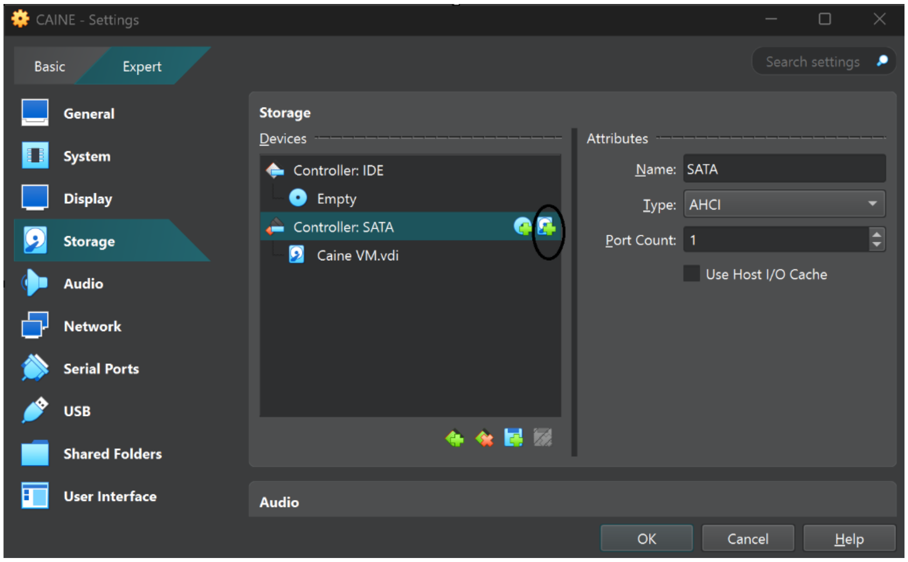


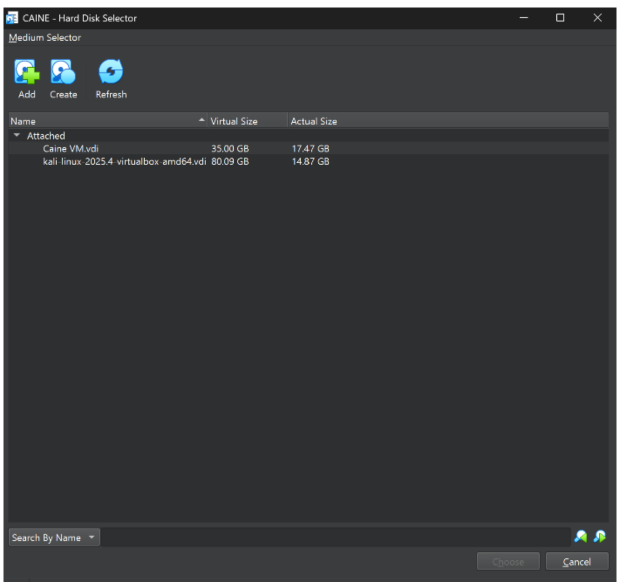


Part 3- Mount virtual disk (.vdi) on CAINE VM and then acquire its image.

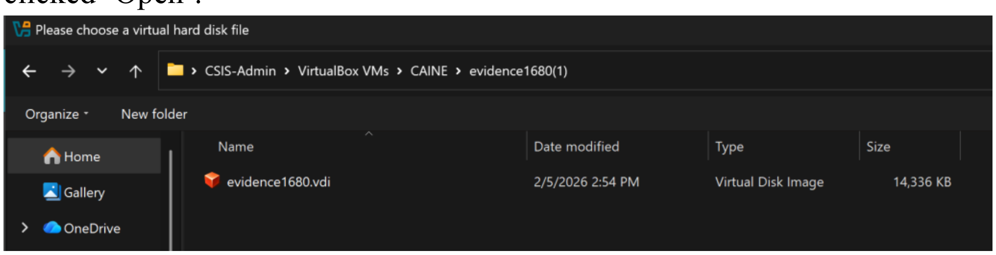
Step 3a - Attaching the evidence disk to CAINE VM

1. Shut down the CAINE VM and clicked on 'Setting' > 'Storage' 'Controller: SATA >'Add Disk.

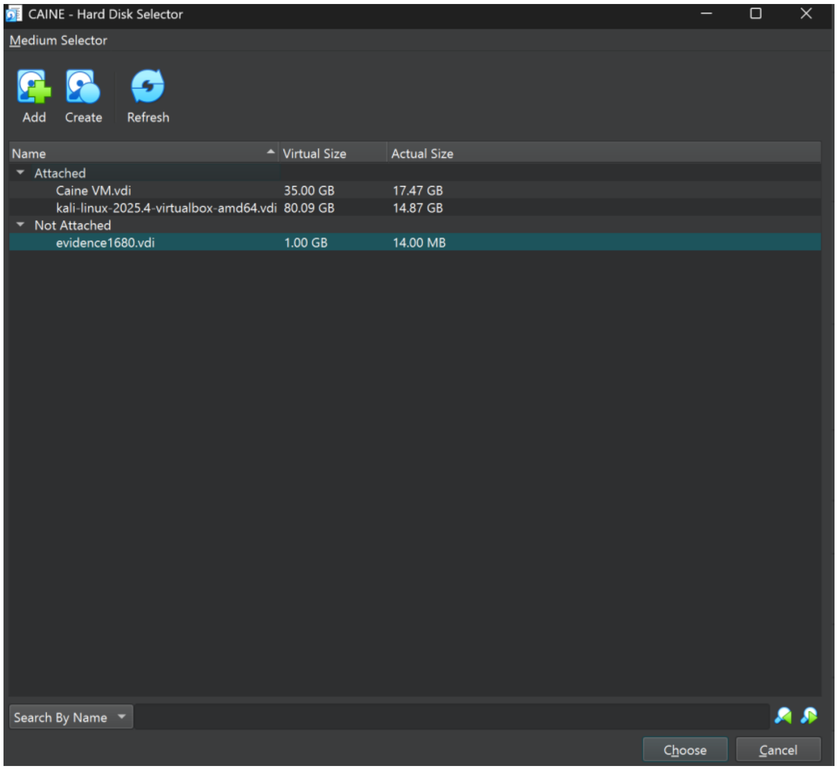




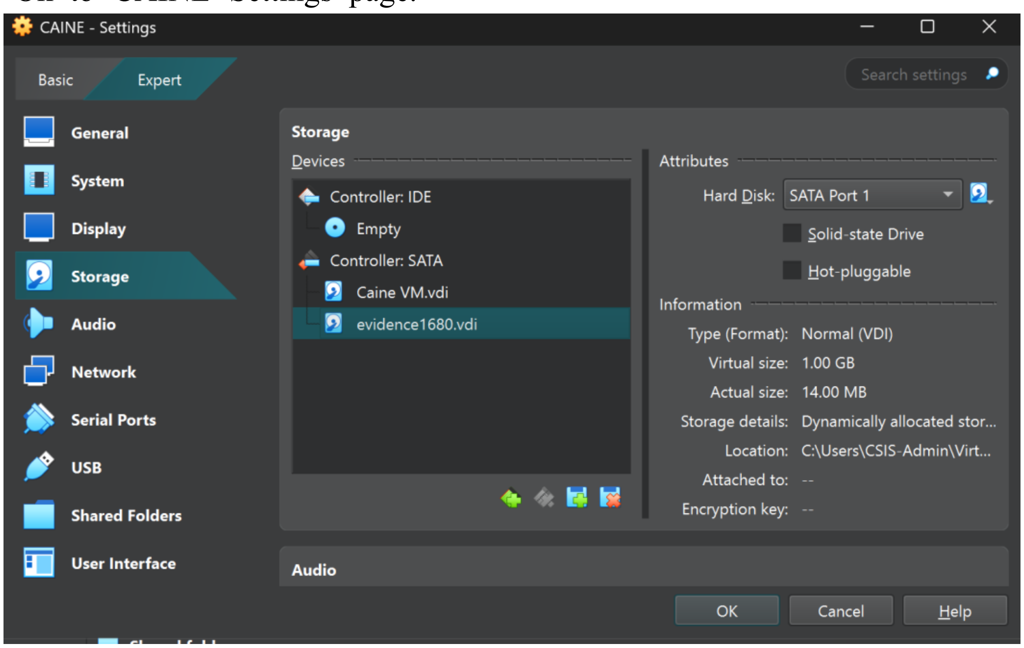
2. Navigated to the folder containing the .vdi file '*evidence1680.vdi*'. Then selected it and clicked 'Open'.



3. Attached the .vdi file. Then selected it and clicked ‘Choose’.

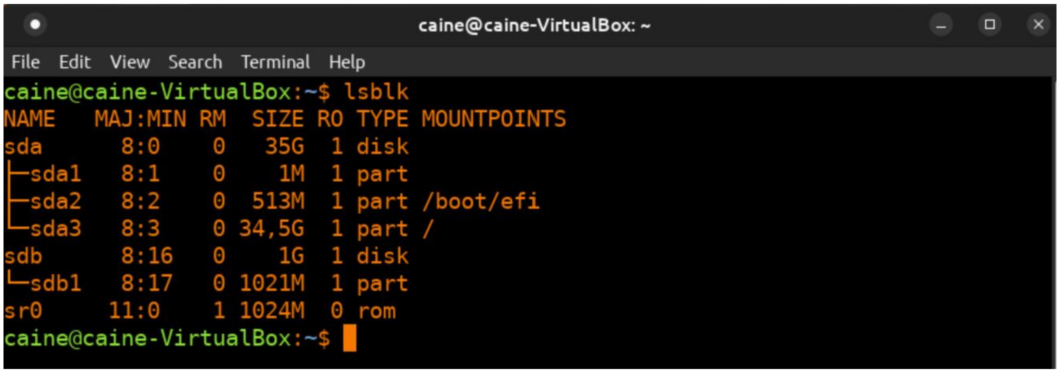


4. The evidence1680.vdi disk attached to the CAINE VM under the SATA controller. Clicked ‘Ok’ to ‘CAINE -Settings’ page.



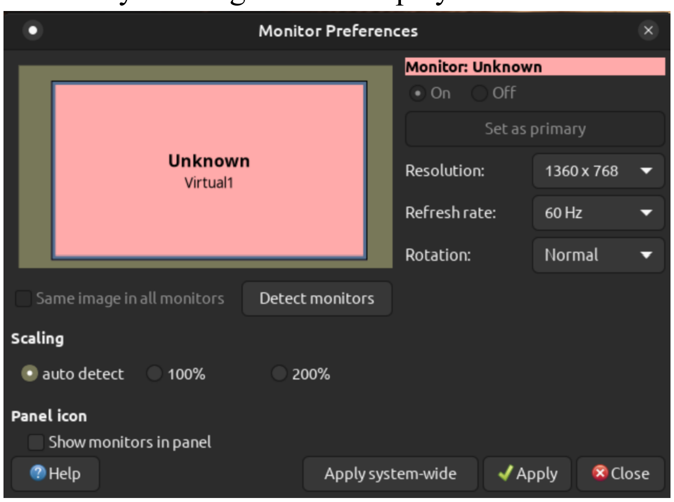
5. After booting the system, the disk appeared as /dev/sdb with one partition (/dev/sdb1) and remained unmounted, ensuring it was not automatically accessed by the system. The RO (read-

only) flag displayed a value of "1", indicating that the device was in read-only mode at the block level.

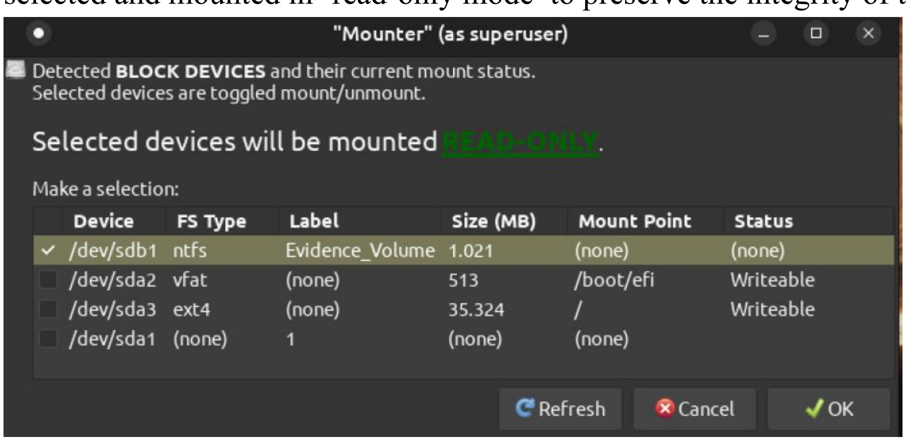


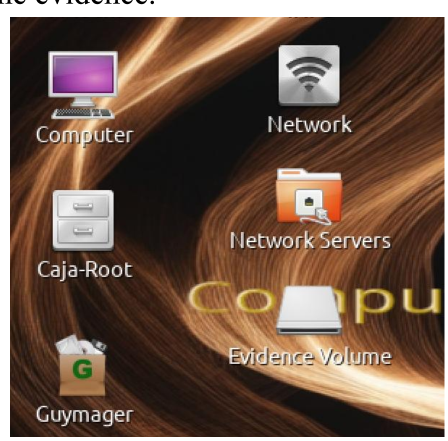
<u>**Step 3b** – Mounting a Disk</u>

1. After booting the system, changed the CAINE VM display settings to see the entire screen at once by clicking on the 'Displays' icon on the desktop and selecting 1360 x768.



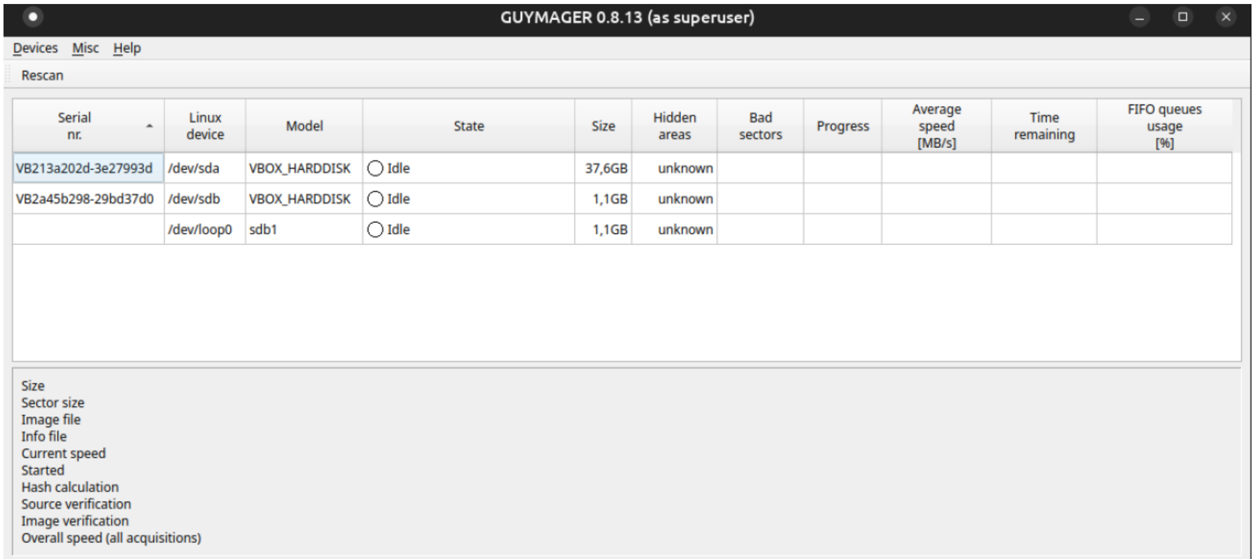
2. The "Safe Devices Mount" tool was then opened, and /dev/sdb1 (the evidence disk) was selected and mounted in 'read-only mode' to preserve the integrity of the evidence.



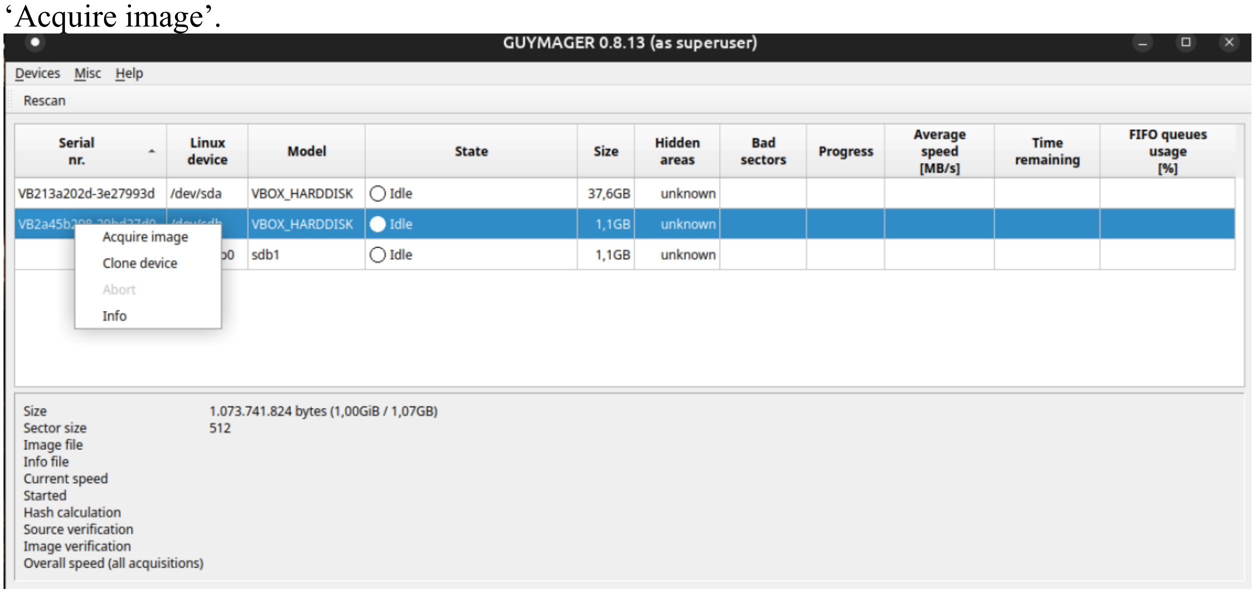


Step 3c - Imaging with Guymager

1. Selected the 'Menu' and opened 'Guymager' under Forensic Tool. (We can also select the application from the Desktop).



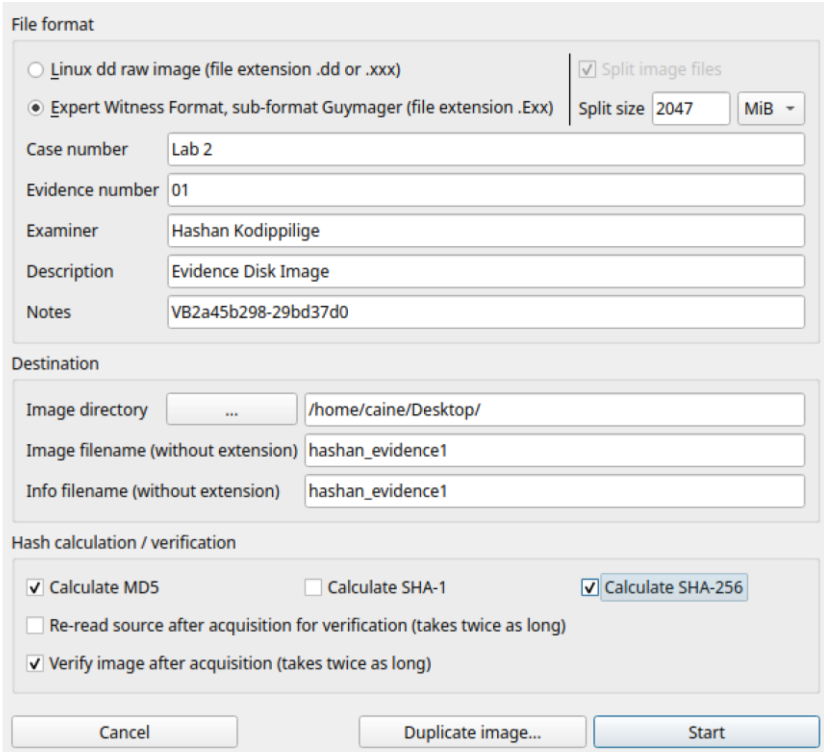
2. The evidence disk /dev/sdb (1.1GB) was identified and selected for acquisition, ensuring the operating system disk (/dev/sda) was not selected. Right clicked on the disk and selected 'Acquire image'.



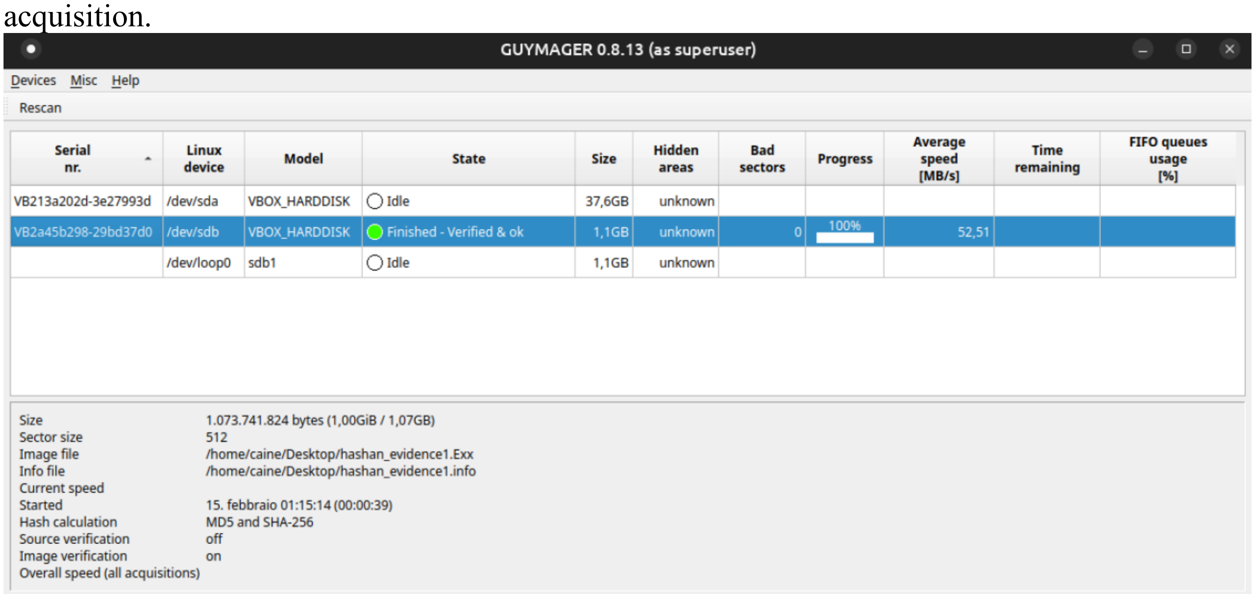
3. In the 'File Format box, following settings were configured.

- File Format: Expert Witness Format (E01)
- Case Number: Lab 2
- Evidence Number: 01
- Examiner: [Your Name]
- Description: Evidence disk image
- Image Directory: /home/caine/Desktop

- Image Filename: yourfirstname_evidence1
- Hash Values: MD5 and SHA-256 enabled



4. The process completed with the status "Finished – Verified & ok," confirming that the forensic image is an exact duplicate of the original evidence disk and that no errors occurred during acquisition.



5. The forensic image file 'hashan_evidence1.E01' and its corresponding metadata file 'hashan_evidence1.info' were compressed into a ZIP folder (hashan_evidence1.zip).

Step 3d - Imaging with dcfldd

1. The evidence disk /dev/sdb was acquired using the dcfldd command-line tool to create a raw forensic image. The following command was used to acquire an image of the evidence disk, sdb.
sudo dcfldd if=/dev/sdb of=/home/caine/Desktop/hashan_evidence2.img hash=sha256 hashlog=/home/caine/Desktop/hashan_hash2.txt status=on

2. The forensic image file 'hashan_evidence2.img' and its corresponding metadata file 'hashan_hash2.txt' were compressed into a ZIP folder (hashan_evidence2.zip).

Questions

1. What important steps did you follow in your imaging procedure to maintain the integrity of the evidence? Discuss.

I have followed few important steps to maintain the integrity of the evidence. I used *lsblk* command to verify that the disk (/dev/sdb) is available on block level and evidence disk was mounted in read-only mode using the Safe Devices Mount tool to prevent any modification of the original data.

When I get the image of the evidence disk using imaging tool Guymager, cryptographic hash values (MD5 and SHA-256) were calculated to verify data integrity. Also, image verification was enabled to confirm that the acquired image matched the original source.

2. Compare the image type you created with Guymager to that you created with the dcfldd tool. Are there any differences? Discuss the implications of this.

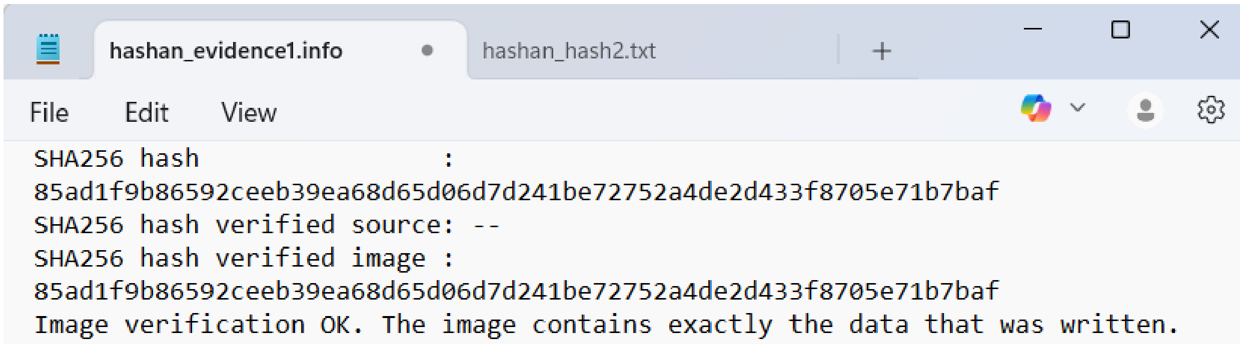
The Guymager created image in the Expert Witness Format (.E01), while dcfldd created with RAW (.img) format. The *.img* format is a "raw" bit-for-bit copy of a disk. Unlike raw images, *.E01* files can be compressed to save space.

The E01 format stores metadata such as case number, examiner name, and hash values within the file. It also supports segmentation and built-in verification. To open .E01 format requires specialized software like FTK Imager or Autopsy, while operating systems (Windows, macOS) can mount *.img* files as virtual drives just by double-clicking them.

When comparing both images, the E01 format is more structured and commonly used in forensic investigations because it preserves additional case information and supports verification features.

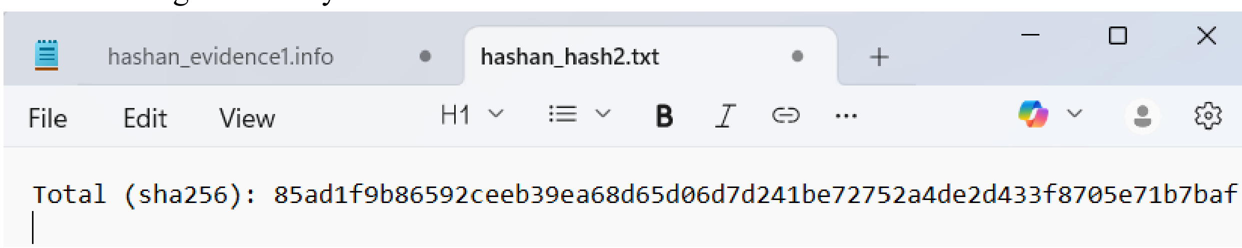
3. Compare and discuss the hash values generated by the Guymager and dcfldd tools.

- Hash value generated by the Guymager tool.




```
SHA256 hash                 :
85ad1f9b86592ceeb39ea68d65d06d7d241be72752a4de2d433f8705e71b7baf
SHA256 hash verified source: --
SHA256 hash verified image :
85ad1f9b86592ceeb39ea68d65d06d7d241be72752a4de2d433f8705e71b7baf
Image verification OK. The image contains exactly the data that was written.
```

- Hash value generated by the dcfldd tool.




```
Total (sha256): 85ad1f9b86592ceeb39ea68d65d06d7d241be72752a4de2d433f8705e71b7baf
```

The SHA-256 hash value produced by Guymager matches the SHA-256 hash generated by dcfldd. This confirms that both forensic tools created identical forensic images of the original evidence disk. It means that data integrity was protected during acquisitions.

If these hash values were different, it would indicate that data may alter during the acquisition or an error during imaging.

Conclusion

This lab helps to cover the practical aspects of forensic imaging process using different digital forensic tools. For this kind of data acquisition need to have CAINE investigative platform. In CAINE investigation flatform, the evidence disk was properly attached, verified, and mounted in read-only mode to prevent any modification of the original data. Two different tools used for imaging processes, which Guymager tool produced an Expert Witness Format (E01) image, and dcfldd tool produced a raw image file (.img).

During acquisition, Guymager generated both MD5 and SHA-256 hash values, and dcfldd generated a SHA-256 hash. The matching SHA-256 hash values confirmed that both images were identical to the original source disk to each other. This ensures that data integrity of evidence was maintained during both imaging processes.

This lab provided hand-on experience in acquiring forensic images while maintaining data integrity. By having two different acquiring methods and matching hash values demonstrated that both imaging methods produced identical copies of the original evidence, which is a critical requirement in digital forensic investigations.

References

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